

Speed of Trade and Liquidity

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Abstract

We investigate whether increasing the speed of order execution affects investor trading strategy and market liquidity. With the new trading platform Arrowhead, the Tokyo Stock Exchange has eliminated the three-second matching cycle, executes orders immediately, and instantaneously updates the limit order book, rendering computerized trading strategies more powerful. Since Arrowhead's introduction, there have been an increase in execution frequency and a reduction in trade size, leading to declines in effective spread and increases in adverse selection costs. Among the trade intensity variables we examine, persistency of runs affects adverse selection cost more than share imbalance. High-speed quote revisions contribute to market depth.

JEL Classification: G10, G12, G14

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1. Introduction

This paper investigates whether increasing the speed of order execution affects trading strategy and liquidity. Recently, global stock exchanges have been competing in how quickly information on quotes and trades can be transmitted. This competition is now in the millisecond to microsecond range because of strong demand from investors who take advantage of technological innovations such as algorithmic trading.

In January 2010, the Tokyo Stock Exchange (TSE) upgraded its trading platform by introducing the Arrowhead system. Many changes were brought about by the new system, a particularly important one involving the matching cycle.¹ The matching interval had been three seconds before the introduction of the new system; now, when a market and/or limit order is entered, it is processed immediately. The order is then executed immediately and the limit order book updated instantaneously. This change has made computerized trading strategies more powerful, because computers can place and cancel orders faster than human traders can. Although algorithmic trading was used in the TSE before 2010, the three-second interval in the matching process prevented quick trading strategies. Trading volume is expected to increase due to the introduction and expanded use of trading algorithms.

Algorithms typically determine the timing, price, and quantity of trades by dynamically monitoring market conditions across different securities, reducing market impact by optimally and sometimes randomly breaking large orders into smaller pieces, and closely

¹ The TSE has revised the rules for maximum price changes between trades and conditions for opening and closing auctions (see TSE, 2010, for further details).

tracking benchmarks such as the volume-weighted average price (VWAP)² over the execution interval.

Faster execution allows liquidity demanders to monitor the market more closely for temporary mispricings or stale quotes, which can raise adverse selection costs (Foucault et al. 2003). Faster execution attracts more informed trading (Barclay and Hendershot 2003). The resulting higher adverse selection can raise the cost of immediacy for liquidity demanders. If investors rely on the same or similar algorithms for their trading decisions, this can increase herding behavior among sophisticated investors (positive feedback effects). The order flow imbalance thus results in a higher transaction cost for liquidity demanders.

Menkveld (2011) focuses on the type of trading referred to as high-frequency trading (HFT). HFT only engages in proprietary trading and starts and ends most trading days with a zero net position. This definition is the same as in such studies as Brogaard (2011) and Kirilenko et al. (2011). Its strategy can be viewed as computerized market making. Fast trading enhances liquidity providers' skills of position risk management and quote setting. It utilizes colocation services³ newly provided by the TSE to maximize the transmission speeds of order submissions, revisions, and cancellations. Orders submitted via colocation servers comprised 10% of total orders in January 2010 and has exceeded 30% since May 2010, according to the TSE. The introduction of HFT in the TSE can help improve stock liquidity.

² The VWAP is used frequently by institutional investors to execute large orders. A VWAP target order refers to the order submission strategy in which a broker slices large institutional orders into small pieces and executes them throughout the day.

³ An exchange allows customers' computer servers to locate on the same floor as the exchange's matching server.

The introduction of Arrowhead in the TSE can provide insights into the tradeoff between the fast order flow from liquidity demanders and competition between liquidity providers. Since the change at the TSE, we have observed an increase in trade frequency and a reduction in trade size, but no significant volume increases. The shift to higher frequency of trades has been pronounced for larger stocks. We estimate a model that captures the relation between quote revisions and trading measures, which we quantify as HFT effects on the quote message traffic (MT) from the new system for individual stocks. Immediate matching promotes competition among liquidity providers, and HFT reduces compensation for liquidity provision while it increases adverse selection costs. Both trends are more pronounced for stocks characterized by heavy HFT, where quote revisions have increased substantially relative to the number of trades since January 2010.

HFT can affect both demand and supply side of liquidity. We select proxies related to either price pressure factor or market-making factor. A proxy for market making is the frequency of quote updates divided by the number of trades, and three order flow proxies are the number of trades (equivalently the trade size), order imbalance, and the persistency of runs. Order imbalances and the persistency of runs indicate increases in market making risk.

Our empirical study shows that quote updates are negatively associated with effective and realized spreads, it means that severe competition between liquidity providers reduces spread measures. Trade size, order (share) imbalance and the persistency of runs are positively correlated with adverse selection. Relation of the persistency of runs with adverse selection is stronger than that of order imbalance. Empirical analysis for extended period in 2010 show

that price pressure and market-making factors becomes more influential than during January 2010.

The remainder of this paper is organized as follows. Chapter 2 explains the TSE's new trading system and briefly surveys related research. Chapter 3 describes our hypotheses and the design of this empirical study. Chapter 4 and 5 discusses our results and the determinants of the spread changes brought about by Arrowhead. Chapter 6 presents our conclusions.

2. The new trading platform

On January 4, 2010, the TSE launched the new trading system Arrowhead. The main features of this system are accelerated computer processing speeds, a colocation service that reduces the physical distance between market participants (investors as well as brokerage firms) and the exchange, and a revision of the tick size. Prior to 2010, the TSE implemented a series of computerizations in the trading process to replace human-based order handling. However, Arrowhead has the potential to incur a major paradigm shift in trading by changing the balance between market participants.

Increasing numbers of papers have begun to address HFT as well as algorithmic trading.

Hendershott and Moulton (2009) study the 2007 introduction of Hybrid Market on the New York Stock Exchange (NYSE). The system expands the use of automated electronic execution and decreases execution times for market orders from over 10 seconds to less than one second. From the month prior to each stock's Hybrid Market activation date to the month after, the NYSE's effective spreads increased from 5.6 to 5.9 basis points due to an increase in adverse

selection. Hendershott et al. (2010) cite the NYSE's 2003 switch to automated quote dissemination to show that algorithmic trading improves liquidity and, equivalently, decreases the speed of price discovery associated with trades. Chordia et al. (2011) note that rapid increases in turnover ratio, a striking increase in the number of trades, and consistent reductions in trade size are characteristic of current US stock markets. These findings reflect the increasing influence of HFT.

High-speed order flow can increase order imbalance for short periods, such as one minute. Chordia and Subrahmanyam (2004) analyze daily order imbalances and indicate that order imbalances can be one of the most important factors to increase the risk of market making in an HFT environment. The authors state that a high absolute order imbalance can alter returns as market makers struggle to readjust their inventory. In addition, order imbalances can signal excessive investor interest in a stock, and if this interest is auto-correlated, then order imbalances can be related to adverse selection costs for liquidity providers.

Auto-correlated order flows can be analyzed by runs tests. Runs tests have been used to test the random walk hypothesis for price movements (Mood., 1940, Fama 1965). Li (2005) estimates the hazard ratio of runs for the directional movement of stock index returns. We apply the runs test to order flow data, where a buyer-initiated trade is assigned a value of +1, and a seller-initiated trade a value of -1. We believe that the run ratio which is the ratio of the number of runs to the number of trades has the potential to capture the persistency of order flow and the associated risk for liquidity providers.

3. Empirical study design

3.1. Hypotheses

Faster execution allows all market participants to monitor market conditions more closely and submit orders more quickly. Thus faster execution attracts more informed investors. The resulting higher adverse selection risk can raise the cost of immediacy for liquidity providers. Quote watchers can replicate trading strategy to follow price movement which generates positive feedback in price formation⁴.

Hypothesis 1: HFT increases adverse selection costs due to more informed and positive feedback traders and faster price discovery.

As trading has become more computerized, it has become easier and cheaper to replace floor traders who played the role of liquidity providers with a computer program. High-frequency liquidity suppliers who monitor market conditions across different securities can quickly notice an abnormally wide bid–ask spread and provide liquidity accordingly via a limit order.

Hypothesis 2: Due to increased competition between liquidity providers, HFT reduces costs of immediacy.

The TSE's introduction of Arrowhead can provide insights into the tradeoff between liquidity demanders' increasing adverse selection and increasing competition among liquidity suppliers.

3.2. Samples

With the TSE's introduction of Arrowhead, the liquidity and price formation of listed stocks may be influenced by two major changes: the matching process cycle and tick size reduction. The former impacts all stocks, while the latter affects only stocks traded within a specific

⁴ Various types of order/execution strategies are described by Harris(1998).

price range, such as ¥2,000 to ¥5,000, as seen Figure 1 for effective spreads and Figure 2 for depth in the best ask and bid price. Since our empirical study focuses on the effect of trading speed, we separate stocks according to whether or not they are affected by the tick size change.

Boosting the speed of trade through Arrowhead is an exogenous change that affects all the stocks listed on the TSE. We collect stocks from the first section of the TSE⁵ and separate them into two groups: the unchanged group, unaffected by tick size changes, and the changed group, affected by tick size changes⁶. The sample period is from one month before the introduction of Arrowhead to one month after. We exclude the last week of December 2009 and the first week of January 2010 based upon our conversations with institutional investors who voiced their reluctance to trade in the brand new system. We exclude stocks that move into and out of different tick sizes or traded for fewer than five days in any month or below ¥100. Table 1 shows a number of stocks in each group and price range. Figure 1 shows the daily movement of the effective spreads and Figure 2 shows the changes in depth for stocks in different price ranges.

(insert Table 1, Figure 1&2 here)

⁵ The TSE has two sections. The listing requirements for the first section of stocks, such as the number of shares outstanding and number of shareholders, are higher than those for the second section.

⁶ Tick size changes affects all spread measures as well as depth in the limit order book. See Goldstein and Kavajecz, 2000, for referece.

4. Arrowhead's impact

4.1. Trading activities

Table 2 summarizes the trading activities for the two groups of stocks, those affected by tick size changes and those not. After Arrowhead's introduction in the TSE, a shift toward high-frequency, small orders was observed in investors' order execution patterns. Of the four trading-related measures selected here, the number of quote revisions shows the largest percentage of change from December 2009 to January 2010. The average number of quote revisions in a five-minute period for the unchanged (changed) group increased from 18.2 (25.6) to 37.1 (73.4). The average number of trades in the five-minute period increased from 5.7 (7.7) to 7.9 (16.6). The average size of trades in the five-minute period decreased from 6.9 (8.8) units to 5.7 (6.6) units. The changes for both groups are statistically significant at the 1% confidence level. The total volume in the five-minute period did not increase for either group.

It should be noted that the transition to HFT is more pronounced for large-cap stocks, and changes in quote revisions, the number of trades, as well as the size of trades are larger. For large-cap stocks, quote revisions in a five-minute period almost triple, from 45.5 (40.3) to 110.8 (125.1) for unchanged (changed) stocks. The average number of trades in the five-minute period increases from 15.0 (12.1) to 23.9 (28.9), and the average size of trades decreases from 22.8 (15.4) units to 16.4 (11.5) units.

An examination of trading-related measures reveals that the changes following Arrowhead's introduction do not occur across the stocks listed in the TSE. Significant changes appear to be concentrated around large-cap stocks. The next section investigates this issue.

(insert Table 2 here)

4.2. Event study of the impact of HFT

Arrowhead's introduction has been characterized by a high frequency of trades and quote revisions. In particular, of the four measures in Table 2, the frequency of quote revisions increases the most. This change is consistent with Hasbrouck and Saar (2009) and Hendershott and Moulton (2008). The change relates to the behavior of high-frequency market making. High-speed transmission improves liquidity providers' ability to change their limit order conditions according to the varying market. In the case of the TSE, elimination of the three-second matching cycle also influence trades and quote MT. Under the three-second matching cycle, market and limit orders that came to market in between were batched and reported cumulatively. Under the Arrowhead system, these orders are executed and reported individually. As a result, the numbers of orders and quotes have increased for high-volume stocks. In addition, investors can enter, change, and cancel orders faster than before the introduction of the new system, thereby affecting their order submission behavior.

To conduct an event study, we quantify these changes by modeling the quote frequency with trade-related variables. An important issue is the normalization of the quote frequency numbers. Hendershott and Moulton (2009) use the number of electronic messages per US\$100 of trading volume as a proxy for algorithmic trading instead of raw MT numbers. We model MT numbers that are influenced by not only the number of trades but also the depth and width of the spread. To estimate the relation for all stocks listed in the TSE, we use four explanatory variables: the *number of trades*, *depth*, *tick spread*, and the *log of the market cap*.

The dependent variable is the number of quote revisions, which is equivalent to MT in Hendershott and Moulton (2009).

First, we estimate the following regression model using daily data from December 2009:

$$MT_{jt} = a + b\#ofTrade_{jt-1} + cDepth_{jt-1} + dTick_spread_{jt-1} + e\log(CapSize_{j0}) + f_{jt} \quad (1)$$

The results in Table 3 show that all variables in equation (1) except tick spread are highly significant and the adjusted R-squared value is 0.88. Using the estimated parameters, we compute the predicted MT for January 2010 and the difference between the predicted and the actual MT (i.e., the prediction error). We assume that the difference indicates the degree of change brought about by HFT, which we hereafter call the HFT effect. We use the model prediction error as a proxy for the HFT effect, which reflects sliced order submissions, changing limit prices as well as quantities, and cancelations. The speedup of a few seconds provides critical new information to high-frequency traders but is unlikely to affect the trading behavior of humans. Elimination of the three-second matching interval allows algorithmic liquidity suppliers to quickly notice abnormally wide inside quotes and provide liquidity accordingly via limit orders. Algorithmic liquidity demanders can quickly access these orders via conventional market or marketable limit orders.

(insert Table 3 & 4 around here)

We form portfolios and sort them into five groups according to HFT effects. In Table 4, the heavy HFT quintile with no tick change (with tick change) stocks shows that the average deviation of the number of quote revisions during five minutes is 56.4 (83.6) from that of the previous month. The smallest HFT quintile shows that the average deviation of the number of

quote revisions is slightly negative, -1.2 (-1.4), which is statistically significantly different from zero. It seems that MT patterns for stocks in lower HFT quintiles do not deviate from those in December 2009. As in Hendershott and Moulton (2009), the results are consistent with the conventional wisdom that algorithmic trading was more prevalent at the time for active, liquid stocks.

4.3. Cost of immediacy

We now examine the effects of Arrowhead's introduction on liquidity. This section focuses on spread measures such as effective spread, realized spread, and adverse selection (market impact) costs. The effective spread is the cost of immediate execution paid to the market by liquidity demanders. The wider the effective spread, the less liquid the stock. For our sample TSE stocks, effective spreads are usually identical to quoted spreads because the TSE uses a purely order-driven mechanism.

For the t th trade in stock j , the proportional effective half-spread, $ESPRD$, is defined as

$$EffectiveSpread(ESPRD)_{jt} = q_{jt} (p_{jt} - m_{jt})/m_{jt}, \quad (2)$$

where q_{jt} is an indicator variable that equals +1 for buyer-initiated trades and -1 for seller-initiated trades, p_{jt} is the trade price, and m_{jt} is the quote midpoint prevailing at the time of the trade. For stock j , each month we calculate the simple average across days and then average it across the month.

Narrower effective spreads imply less revenue per trade for liquidity providers. We decompose effective spreads into a realized spread component ($RSPRD$) and an adverse

selection or price impact component (MI) to understand the source of the improvement in liquidity under Arrowhead's implementation (Glosten 1987):

$$ESPRD_{jt} = RSPRD_{jt} + MI_{jt} = (q_{jt} (p_{jt} - m_{j,t+5min}) / m_{jt}) + (q_{jt} (m_{j,t+5min} - m_{jt}) / m_{jt}), \quad (3)$$

where p_{jt} is the trade price, q_{jt} is the buy–sell indicator (+1 for buys and –1 for sells), m_{jt} is the midpoint prevailing at the time of the t th trade, and $m_{j,t+5min}$ is the quote midpoint five minutes after the t th trade. We estimate the revenue to liquidity providers using the five-minute realized spread, which assumes the liquidity provider is able to close his or her position at the quote midpoint five minutes after the trade. We measure gross losses to liquidity demanders due to adverse selection using the five-minute market impact of a trade.

Table 5 and Figure 3 compares these measures between December 2009 and January 2010. Overall changes in effective spreads are negative for all quintiles. Stocks in the largest HFT quintile show the smallest decline (–0.006%), and those in the two lighter quintiles show a larger decline (–0.031%, –0.039%). The heavier the HFT effect, the smaller the reduction in the cost of immediacy. It seems that high-speed transactions somewhat prevent the reduction of the cost of immediacy.

Decomposing the effective spread into liquidity provider revenues—the realized spread—and adverse selection—the permanent market impact—we find effects in two opposite directions. For stocks in a heavy HFT quintile, the realized spread shows a statistically significant decline (–0.014%) and market impact shows a statistically significant increase (0.008%), but for stocks in a light HFT quintile, the realized spread significantly increases (0.017%) and market impact significantly declines (–0.048%). From the results above, compensation for liquidity providers is reduced in the five-minute period due to positive market impact. This

finding contradicts the results of Hendershott et al. (2010) in the US market, where both the effective spread and market impact declined. For stocks in smaller HFT quintiles, however, the source is the reduction of permanent market impact costs, similar to US stocks.

(insert Table 5 and Figure 3 around here)

4.4. Determinants of execution costs

We have observed a reduction in the cost of immediacy and compensation for liquidity provision. We now investigate the determinants of these changes. Candidate variables are the HFT effect, the cumulative yen volume, the number of trades, and the size of trades. The natural way to test these variables is by regressing the various liquidity measures, L_i , on HFT_i , Vol_i , $\#Trade_i$, and $TradeSize_i$ and controlling variables X_i :

$$\Delta L_i = \alpha + \beta_1 HFT_i + \beta_2 \frac{Vol_{i,Jan}}{Vol_{i,Dec}} + \beta_3 \frac{\#Trade_{i,Jan}}{\#Trade_{i,Dec}} + \beta_4 \frac{TradeSize_{i,Jan}}{TradeSize_{i,Dec}} + \gamma \Delta X_j + \varepsilon_i, \quad (4)$$

where ΔL_i is $\Delta ESPRD_i = ESPRD_{i,Jan} - ESPRD_{i,Dec}$, differences in the effective spread between December 2009 and January 2010; similarly, $\Delta RSPRD_i$ is the realized spread and ΔMI_i the market impact; HFT_i is the prediction error of equation (1); $Vol_{i,Jan} / Vol_{i,Dec}$ is the ratio of the yen volume on January 2010 to that on December 2009; similarly, $\#Trade_{i,Jan} / \#Trade_{i,Dec}$ is the ratio of the numbers of trades, $\#TradeSize_{i,Jan} / \#TradeSize_{i,Dec}$ is the ratio of trade sizes, and the X_i are stock-level control variables, including $\Delta tickspread$, the difference in tick spread, and $\Delta tickspread \times tick\ size\ change\ dummy$, which is the difference in tick spread for stocks affected by tick size change, and zero otherwise.

Table 6 presents the coefficients for variants of equation (4). When the dependent variable L_i is $\Delta ESPRD$ for stock i , the coefficients of HFT_i are positive and marginally significant at the 10% level. This means that the larger the HFT effect, the larger the increase in effective spread. The variables of the ratios of $\#Trade$ and $TradeSize$ show significant negative relations with effective spread change. This means that the larger the increase in the number of trades or in the size of the trades, the larger the reduction in effective spread. It is interesting that the ratio of Vol has a significant, positive coefficient with effective spread change. Here, Vol is a product of the number of trades and their size, so increasing volume widens the effective spread due to adverse selection risk. The results indicate that the changes in effective spread are more strongly related to the number of trades and trade size than the HFT effects.

When the dependent variable L_i is $\Delta RSPRD_i$ for stock i , the coefficients of HFT are negative and significant at the 1% level. The variables $\Delta \#Trade$ and $\Delta SizeTrade$ do not have significant coefficients. This means that the larger the HFT effect, the greater the reduction in compensation for liquidity providers.

When the dependent variable L_i is ΔMI_i for stock i , the coefficient of HFT is positive and significant at the 1% level. Frequent quote revisions are related to a permanent price impact after the trade. Here, $\Delta \#Trade$ and $\Delta SizeTrade$ have a significant negative relation, which means that the larger the HFT effect, the greater the market impact. The negative relations of $\Delta \#Trade$ and $\Delta SizeTrade$ imply that the frequency and size of trades indicate high liquidity.

From the results in Table 6, we conclude that the overall reduction in effective spread is due to the increased number of trades as well as the size of trades. Increased MT intensifies adverse selection risks and makes a permanent market impact after the trade.

(insert Table 6 around here)

5. Extended sample periods and additional tests

To perform robustness checks, we extend the sample period by including October, November, and December 2010. In the event study in the previous section, January 2010 represents a month of the TSE's new trading platform; however, participants may need time to adjust their trading parameters. High-frequency traders rely on computerized trading strategies and want to test the performance of their strategies with historical data under the new environment before they start trading. According to the TSE, orders via colocation servers comprised about 10% of total orders in January 2010 and have exceeded 30% since May 2010. This indicates more high-speed participants became ready for trading after May 2010. In addition, the share of off-exchange markets began to increase after October 2010. It is said that high-frequency market makers trade in multiple trading venues (Menkveld 2011, Brogaard 2011). It seems that the additional three months of data provide more reliable results after 2010.

5.1. Summary statistics for liquidity measures

Table 7 shows the summary statistics for spreads and depth measures for a five-month period December 2009, January, October, November and December 2010. The effective spread gradually declines closer to December 2010. As of December 2009, the average effective

spread is 0.20% and it does not change much until November 2010. As of December 2010, the average effective spread is 0.16%, an almost 20% reduction. The realized spread for December 2010 is 0.060%, which is slightly higher than for December 2009 and January 2010 (0.052% and 0.52%, respectively). The adverse selection cost for December 2010 is 0.10%, which is smaller than for December 2009 and January 2010 (0.16% and 0.14%, respectively). The average depth at the best ask and bid prices in the limit order book for December 2010 is 99,235 shares, which is almost double that for December 2009 and January 2010 (51,386 and 55,262, respectively). In summary, participants have adapted to the new trading environment. They mitigate market impact costs, which is equivalent to adverse selection in our definition, which results in a smaller cost of immediacy. The increased depth may indicate that the market-making of high-frequency traders becomes more active as the off-exchange volume increases.

(insert Table 7 around here)

5.2. Price pressure and market-making

High-speed trading can be the cause of two countervailing effects on liquidity: liquidity taking and liquidity provision. If orders are more concentrated on one side, a higher degree of order imbalance deteriorates market liquidity. Increasing the speed of trading amplifies order imbalance, thus having a permanent market impact after a trade. We investigate this phenomenon by introducing three trading intensity variables such as share imbalance and the persistency of runs, in addition to the number of trades (or the size of trades).

First, the number of trades or the size of trades are an activity measure (Jones et al., 1994). The next variable is share imbalance, which is the difference between the number of buyer- and seller-initiated shares divided by total volume over a 30-minute period (Chordia and Subrahmanyam, 2004). In the TSE, which is a purely order-driven market, we can determine the initiating side of a trade by comparing the prevailing quotes with the price of the following trade. The elimination of the three-second matching interval affects the count of the number of trades, so a share volume-based imbalance is more appropriate for use in this study. The third variable is the standardized length of a sequence based upon the runs test. The runs test often applies to price changes rather than order flow. Here we assign +1 for a buyer-initiated order and -1 for a seller-initiated order.

Based upon Fama (1965), we compute the predicted value for the run ratio given a number of trades and the order imbalance ratio between buyer- and seller-initiated orders⁷. We then compute the deviation of actual run ratios from the predicted ratio divided by the standard error of the run ratio⁸. This measure captures the persistency of one-sided order flow for a short period. From the market-making point of view, a sequence of buy (or sell) trades monotonically decreases(increases) inventory and may have an impact on market-making behavior. All three variables uniquely relate to the intensiveness of order flow. For the liquidity provision side, a variable related to market making is the frequency of quote updates normalized by the number of trades.

⁷ Under the assumption of independence the total expected number of runs of all signs for a Stock (m) can be computed as $m = [N(N + 1) - \sum n^2]/N$, where N is the total number of trades, and the n are the numbers of (buy/sell) side changes of each orders. See equation 13 in Fama (1965).

⁸ Computation of standard error of predicted run(m in footnote 7) is given by equation 14 in Fama (1965)

5.3. Determinants of liquidity

This section tests whether the level of liquidity as well as its relation with trading measures changes between December 2009 and January 2010 (referred to as the first period) and October, November, and December 2010 (referred to as the second period) by the following regression analysis. The basic form of the regression equations is

$$L_{i,t} = \alpha + \beta_1 \frac{QuoteFreq_{i,t}}{TradeFreq_{i,t}} + \beta_2 TradeSize_{i,t} + \beta_3 ShareImb_{i,t} + \beta_4 RunPersistency_{i,t} + \gamma \Delta X_j + \varepsilon_t, \quad (5)$$

where L_i represents four types of liquidity measures: $ESPRD_i$, the effective spread; $RSPRD_i$, the realized spread; MI_i , the adverse selection cost; and $Dpth_i$, the average depth at the best ask and bid prices. Our market-making proxy is $QuoteFreq_{i,t}/TradeFreq_{i,t}$, the normalized quote update frequency. We have three order flow intensity proxies: $TradeSize_i$, $ShareImb_i$, and $RunPersistency_i$. The X_i are stock-level control variables, including the logarithmic market capitalization of stock, a tick size change dummy that equals one for stocks subject to tick size change and zero for other stocks, and dummies for the time of day (by 30-minute intervals). The time dummy for 10:00 A.M. is not included in the regression.

To test the changes, we insert two period dummies: one for December 2009 that equals one for observations for December 2009 and zero otherwise, and a period dummy that equals one for observations for October to December 2010 and zero otherwise. These dummies are inserted to detect changes in the intercept and the coefficients of $QuoteFreq_{i,t}/TradeFreq_{i,t}$, $TradeSize_i$, $ShareImb_i$, and $RunPersistency_i$.

(insert Table 8 around here)

Table 8 shows the results of regressions. From the intercept and the coefficients of its dummies, the effective half-spread declines by 6.2 basis points on January 2010 compared to December 2009. During the second period of 2010 there is a further reduction of 1.9 basis points. The realized spread and adverse selection cost show smaller changes but similar trends.

The market-making factor (quote update) shows a negative coefficient for effective spread: $-0.0053 (= -0.0010 - 0.0043)$ for December 2009, -0.001 for January 2010, and it remains almost the same for the second period of 2010. The quote update shows a similar trend for the realized spread and adverse selection cost. The change between the first and second period in 2010 is insignificant. Quote updates relate to a reduction in the cost of immediacy and market-making profits.

Among the order flow intensity proxies, trade size has a positive relation with effective spread, realized spread, and adverse selection cost. The positive relation with the three spread measures remains the same after 2010. This means that the reduction in average trade size after 2010 contributes to a reduction in effective spread. Order imbalance has a positive relation with effective spread and adverse selection cost, which is consistent with Chordia and Subrahmanyam (2004). These authors state that order imbalances may be related to adverse selection cost for liquidity providers. The persistency of runs is also positively correlated with adverse selection cost, but negatively correlated with effective spread and realized spread. This means that runs lasts longer than theoretically predicted when competition among liquidity providers is intense.

The average depth of the bid–ask price in the limit order book is positively correlated with quote updates, trade size, and order imbalance. The only factor that is negatively correlated with depth is the persistency of runs. This finding is consistent with the notion that the length of a sequence on the one side of order is a risk for liquidity providers. We confirm this relation by the fact the persistency of runs is also positively correlated with adverse selection cost. The larger the deviation of the run ratio from the predicted value, the lower the depth.

The degree of impact generated by explanatory variables cannot be assessed from the size of their coefficients because their means change before/after the Arrowhead introduction. To illustrate the changes brought about by the new trading system, we compute the value of four components: market making (quote update) and order flow intensity (trade size, order imbalance, and the persistency of runs). Each amount is computed by multiplying the estimated coefficients and mean values of the explanatory variables for each period.

Figure 4 illustrates the size and side of the impact for the four variables. Sample stocks are those categorized as a heavy HFT influence in Section 4.2. Spread reduction by market-making activities attenuates after 2010, and higher-frequency and smaller trades gradually increase the three spread measures. The persistency of runs affects adverse selection cost more than share imbalance, and it reduces the compensation of liquidity providers. Its effects increase in the new trading platform. For depth, market-making factors become increasingly important, and the persistency of runs is a negative factor.

(insert Figure 4 around here)

6. Concluding remarks

Since Arrowhead's introduction at the TSE, higher-frequency and smaller trades have been observed without a volume increase. Of the transaction-related measures, the number of quote revisions increases the most. We estimate a model that captures the relations between quote updates and transactions and quantify an HFT effect for individual stocks in the new trading platform.

The US Securities and Exchange Commission (2010) expressed concern over the HFT effect and whether it improves liquidity or diminishes price efficiency through abrupt price movements up and down. After the launch of Arrowhead in the TSE, the effective spread declined and the compensation for liquidity provision decreased due to increased adverse selection costs. Both of these trends are more pronounced for large-cap stocks characterized by HFT.

The TSE's new trading platform significantly enhanced order execution and turnaround. The three-second matching cycle was eliminated and the dissemination of trades and quotes is now carried out instantaneously and individually. We compare the liquidity in December 2009 and in January, October, November, and December 2010 and determine the new trading platform's impact on liquidity. The increased number of trades may be the natural result of more frequent matching. Reporting trades and quote revisions individually should have a real impact on investor behavior.

Some market participants at the TSE say that quote changes have become too fast to be perceived by humans. This indicates that, in the new trading platform, computerized market-making replaces human-based activity and leads to intense competition among liquidity providers. This has had a negative impact on the realized spread.

Higher adverse selection costs may imply that increasing the speed of trade amplifies order imbalance, thus having a permanent market impact after the trade. We investigate this by introducing trading intensity variables such as share imbalance and the persistency of runs, in addition to the number of trades.

We find that spread reduction by market-making activities attenuates after 2010 and higher-frequency and smaller trades gradually increase the cost of immediacy. The persistency of runs affects adverse selection cost more than share imbalance and increases the risk of market making. For depth, market-making factors become increasingly important, and HFT contributes to market depth later in 2010.

Brokerages engaging in short-term trading for their own accounts will have difficulty avoiding reduced profitability due to increasingly intense competition with HFT, making it a challenge to remain viable without some form of HFT capability. The use of HFT remains unchanged, however, for small-cap stocks, a segment of the market where algorithmic trades cannot be said to have increased the supply of liquidity.

Several important questions remain unanswered, due largely to a lack of detailed data. Beyond the observed changes, it would be interesting to determine which investors change

their investment behavior and in what manner. A growing off-exchange trading may be connected with the liquidity improvement we observe in the later months of 2010. We will continue such investigations in future projects.

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Table 1. Sample stocks and tick size.

We collect stocks from the TSE one month before and one month after the introduction of the Arrowhead system on January 4, 2010. We exclude the last week of December and the first week of January. As of January 4, 2010, the TSE revised the tick size table, so our sample stocks must remain within the same price range listed below during the entire sample period. We exclude stocks that traded for fewer than five days in any month and below ¥100.

Price range	Number of stocks	old tick	new tick
Below 2000	1232	¥1	same
2000~3000	46	¥5	¥1
3000~5000	35	¥10	¥5
5,000~30,000	53	¥10	same
30,000~50,000	10	¥50	same
50,000~300,000	61	¥100	same
300,000~500,000	12	¥1,000	¥500
500,000~1,000,000	8	¥1,000	same
Total	1457		

Table 2. Trading measures on December 2009 and January 2010.

This table examines the number of trades, their size, cumulative volume, number of quote revisions, and number of observations. All measures are averaged over five-minute intervals each day, over the month, and then over the stocks. Stocks are categorized into two groups, one with no changes in tick size and the other with changes in tick size. Within each group, stocks are sorted by market cap to form quintiles. Test statistics are computed as $t = (\bar{x}_j - \bar{x}_d) / \sqrt{s_j^2/n_j + s_d^2/n_d}$, where the means of each measure for December and January are $\bar{x}_d$ and $\bar{x}_j$, respectively; the standard deviations are s_d and s_j , respectively; and the numbers of observations are n_d and n_j , respectively.

Tick size	Large	2	Mid	4	small	all	Large	2	Mid	4	small	all	Large	2	Mid	4	small	all
	Dec-09						Jan-10						test stats					
Number of Trades																		
unchanged	15.0	7.2	4.0	2.3	1.4	5.7	23.9	9.0	4.5	2.6	1.9	7.9	8.6	3.5	2.8	2.0	2.6	9.4
changed	12.1	4.9	2.1	1.9	2.3	7.7	28.9	7.2	2.6	2.0	4.7	16.6	5.5	3.4	2.2	0.6	–	5.3
all	14.6	6.9	4.0	2.3	1.4	5.8	24.6	8.8	4.4	2.6	1.9	8.5	10.1	4.0	2.9	2.0	2.6	10.6
Size of Trade																		
unchanged	22.8	4.6	3.0	2.6	3.5	6.9	16.4	4.0	3.1	2.5	3.8	5.7	–2.0	–2.7	0.2	–1.2	0.8	–2.1
changed	15.4	3.1	1.7	2.8	9.5	8.8	11.5	2.2	1.5	2.6	9.4	6.6	–7.0	–3.2	–1.5	–1.4	–	–6.7
all	21.7	4.5	3.0	2.6	3.6	7.1	15.6	3.9	3.0	2.5	3.8	5.8	–2.2	–3.2	0.2	–1.3	0.8	–2.3
Cumulative Volume in five minute																		
unchanged	444.5	63.4	21.3	14.3	10.1	102.3	452.7	72.8	26.5	15.8	14.7	107.9	0.3	1.1	1.2	0.7	1.7	1.0
changed	187.2	23.6	5.4	10.7	30.0	98.2	237.4	21.6	5.4	9.2	48.9	121.3	1.7	–1.2	0.1	–0.9	–	1.6
all	405.6	59.1	20.8	14.2	10.2	102.0	420.1	67.4	25.7	15.7	14.8	108.7	0.6	1.1	1.2	0.7	1.7	1.2
Quote Revisions																		
unchanged	45.5	23.5	13.6	7.9	5.0	18.2	110.8	42.6	21.2	12.5	8.7	37.1	14.1	10.3	9.3	6.4	7.1	16.7
changed	40.1	15.8	7.1	6.4	6.0	25.6	125.1	35.0	14.1	11.7	15.3	73.4	7.5	4.9	3.5	2.2	–	7.2
all	44.7	22.7	13.4	7.9	5.0	18.7	113.0	41.7	20.9	12.5	8.7	39.4	15.9	11.2	9.6	6.6	7.2	18.0
Number of observations																		
unchanged	247	260	281	284	290	1,362	247	260	281	284	290	1,362	–	–	–	–	–	–
changes	44	31	10	7	1	93	44	31	10	7	1	93	–	–	–	–	–	–
all	291	291	291	291	291	1,455	291	291	291	291	291	1,455	–	–	–	–	–	–

Table 3. MT model estimation.

Using daily data for December 2009, we estimate the regression model for MT:

$$MT_t = a + b\#ofTrade_{t-1} + cDepth_{t-1} + dTick_spread_{t-1} + e\log(CapSize_0) + f_t$$

The explanatory variables are *number of trades*, *depth*, *tick_spread*, and *log_marketcap*. The dependent variable is the frequency of quote revisions (*MT*). To avoid the problem of endogeneity, we use the lagged values of the first three explanatory variables and set *log_marketcap* as of the end of November.

Variables	Coefficient	Standard Error	t-value
Intercept	-38.208	0.669	-57.12
Depth(-1)	-6.773×10 ⁻⁷	1.889×10 ⁻⁷	-3.59
#Trade(-1)	2.115	0.008	265.73
Tspread(-1)	0.013	0.014	0.91
LogCap	2.522	0.034	63.91
Adj.R-sqr	0.881		
observations	23,470		

Table 4. Forecasting error of the MT model.

Using the estimated parameters in Table 3, we compute the predicted MT for January 2010 and the difference between the predicted and the actual MT (i.e., the prediction error). We assume that the difference indicates the degree of change brought about by the HFT, that is, the HFT effect.

		HFT effect					
		Light	2	Mid	4	Heavy	All
no tick change	Mean	-1.19	1.40	5.21	14.31	56.43	15.26
	St. Dev	1.31	0.67	1.68	4.43	38.03	27.32
	Obs.	272	272	272	273	273	1,362
	t-value	-15.0	34.7	51.0	53.4	24.5	20.6
tick change	Mean	-1.37	6.47	17.67	38.13	83.56	30.66
	St. Dev	2.16	3.78	3.98	8.80	39.56	36.86
	Obs.	18	18	18	18	21	93
	t-value	-2.7	7.3	18.8	18.4	9.7	8.0

Table 5. Spread measures.

The effective spread is the cost of immediate execution paid to the market by liquidity demanders. We decompose effective spreads into a realized spread component and an adverse selection or price impact component (see equation (3)).

Panel A: Stocks not affected by tick size changes

	HFT Effect	Average	Difference	St.Dev	Sample	t-value
Effective Spread	Light	0.267	-0.031	0.129	272	-3.9
	2	0.257	-0.039	0.151	272	-4.3
	3	0.204	-0.018	0.072	272	-4.2
	4	0.140	-0.006	0.050	273	-2.1
	Heavy	0.099	-0.006	0.016	273	-6.1
Realized Spread	Light	0.079	0.017	0.144	272	2.0
	2	0.054	0.015	0.143	272	1.7
	3	0.032	0.010	0.087	272	1.8
	4	0.035	-0.001	0.080	273	-0.2
	Heavy	0.031	-0.014	0.023	273	-10.4
Adverse Selection	Light	0.187	-0.048	0.111	272	-7.1
	2	0.203	-0.054	0.141	272	-6.3
	3	0.172	-0.028	0.088	272	-5.2
	4	0.105	-0.005	0.073	273	-1.2
	Heavy	0.068	0.008	0.020	273	7.0

Note: Difference=Lj-Ld

Panel B: Stocks affected by tick size changes

	HFT Effect	Average	Difference	St.Dev	Sample	t-value
Effective Spread	Light	0.181	-0.052	0.091	18	-2.4
	2	0.091	-0.057	0.023	18	-10.7
	3	0.080	-0.055	0.016	18	-14.6
	4	0.066	-0.062	0.011	18	-23.6
	Heavy	0.045	-0.072	0.010	21	-31.3
Realized Spread	Light	0.036	-0.001	0.098	18	-0.1
	2	0.007	-0.031	0.016	18	-8.0
	3	0.003	-0.033	0.021	18	-6.7
	4	0.007	-0.045	0.037	18	-5.2
	Heavy	0.013	-0.067	0.016	21	-19.2
Adverse Selection	Light	0.146	-0.051	0.060	18	-3.6
	2	0.083	-0.026	0.032	18	-3.5
	3	0.077	-0.022	0.026	18	-3.7
	4	0.059	-0.017	0.030	18	-2.4
	Heavy	0.033	-0.005	0.011	21	-2.1

Table 6. Determinants of spread change.

The following regression equation is estimated:

$$\Delta L_i = \alpha + \beta_1 HFT_i + \beta_2 \frac{YenVol_{i,Jan}}{YenVol_{i,Dec}} + \beta_3 \frac{\#Trade_{i,Jan}}{\#Trade_{i,Dec}} + \beta_4 \frac{TradeSize_{i,Jan}}{TradeSize_{i,Dec}} + \gamma \Delta X_j + \varepsilon_i,$$

where ΔL_i is $\Delta ESPRD_i = ESPRD_{i,Jan} - ESPRD_{i,Dec}$, the difference in the effective spread between December 2009 and January 2010; $\Delta RSPRD_i$ (realized spread) and ΔMI_i (market impact) are similarly defined; HFT_i is a prediction error of the equation in Table 3; $YenVol_{i,Jan} / YenVol_{i,Dec}$ is the ratio of the yen volume on January 2010 to that on December 2009; similarly, $\#Trade_{i,Jan} / \#Trade_{i,Dec}$ is the ratio of the numbers of trades; $TradeSize_{i,Jan} / TradeSize_{i,Dec}$ is the ratio of trade sizes; and the X_i are stock-level control variables, including $\Delta tickspread$, the difference in tick spread, and $\Delta tickspread \times tick\ size\ change\ dummy$, the difference in tick spread for stocks affected by tick size change, and zero otherwise.

	$\Delta ESPRD$		$\Delta RSPRD$		ΔMI	
Variable	Coefficient	t-Statistic	Coefficient	t-Statistic	Coefficient	t-Statistic
<i>ArrowheadEffect/100</i>	0.0125	1.82	-0.0407	-3.98	0.0532	5.49
ΔVol	0.0054	2.42	-0.0027	-0.81	0.0081	2.58
$\Delta TradeSize$	-0.0272	-2.86	0.0157	1.11	-0.0429	-3.21
$\Delta NumberofTrade$	-0.0157	-3.96	-0.0038	-0.65	-0.0119	-2.13
$\Delta TickSpread$	0.0535	41.33	0.0316	16.39	0.0220	12.04
<i>TicChangeDummyX</i> $\Delta TickSpread$	-0.0318	-8.94	-0.0165	-3.12	-0.0153	-3.05
$\Delta Depth$	0.0010	0.55	0.0029	1.08	-0.0019	-0.76
<i>Intercept</i>	0.0228	2.53	0.0046	0.35	0.0182	1.43
Adjusted R-squared	0.561		0.173		0.129	
Observation	1,455		1,455		1,455	

Note: The t -statistics are White heteroskedasticity-consistent standard errors and covariance.

Table 7. Liquidity changes for the extended periods.

A summary of the four liquidity measures effective spread, realized spread, adverse selection cost, and average depth for December 2009, and January, October, November, and December 2010.

Stocks subject to tick size changes in January 2010 are excluded.

	Effective Spread				
	Dec 2009	Jan 2010	Oct 2010	Nov 2010	Dec 2010
Mean	0.2043	0.1908	0.1911	0.1841	0.1627
Median	0.1710	0.1597	0.1564	0.1474	0.1303
Std. Dev.	0.1386	0.1233	0.1355	0.1295	0.1228
Observations	1324	1324	1324	1324	1324
	Realized Spread				
	Dec 2009	Jan 2010	Oct 2010	Nov 2010	Dec 2010
Mean	0.0517	0.0524	0.0479	0.0532	0.0598
Median	0.0312	0.0314	0.0254	0.0303	0.0337
Std. Dev.	0.1051	0.0945	0.0938	0.0884	0.1150
Observations	1322	1322	1322	1322	1322
	Adverse Selection Cost				
	Dec 2009	Jan 2010	Oct 2010	Nov 2010	Dec 2010
Mean	0.1564	0.1393	0.1433	0.1309	0.1035
Median	0.1207	0.1098	0.1171	0.1017	0.0848
Std. Dev.	0.1379	0.1216	0.1050	0.1032	0.0773
Observations	1322	1322	1322	1322	1322
	Depth				
	Dec 2009	Jan 2010	Oct 2010	Nov 2010	Dec 2010
Mean	51,386	55,262	76,264	82,368	99,235
Median	4,314	4,610	4,899	4,979	5,585
Std. Dev.	417,561	421,573	840,432	967,914	1,230,798
Observations	1324	1324	1324	1324	1324

Table 8. Determinants of spread measures.

The basic form of regression equations are as follows:

$$L_{i,t} = \alpha + \beta_1 \frac{QuoteFreq_{i,t}}{TradeFreq_{i,t}} + \beta_2 TradeSize_{i,t} + \beta_3 ShareImb_{i,t} + \beta_4 RunPersistency_{i,t} + \gamma \Delta X_j + \varepsilon_i$$

where L_i represents four types of liquidity measures, such as $ESPRD_i$, the effective spread; $RSPRD_i$, the realized spread; MI_i , adverse selection cost; and $Dpth_i$, the average depth for the best ask and bid prices. Our market-making proxy is $QuoteFreq_{i,t}/TradeFreq_{i,t}$, the normalized quote update frequency. We have three order flow intensity proxies: $TradeSize_i$, $ShareImb_i$, and $RunPersistency_i$. The X_i are stock-level control variables, including the logarithmic market capitalization of stock, a tick size change dummy that equals one for stocks subject to tick size changes and zero for other stocks, and dummies for time of day (at 30-minute intervals). The time dummy of 10:00 A.M. is not included in the regression.

Variable	Effective Spread		Realized Spread		Adverse Selection		Depth	
	Coefficient	t-Statistic	Coefficient	t-Statistic	Coefficient	t-Statistic	Coefficient	t-Statistic
Intercept								
C	0.2404	264.1	0.0389	21.8	0.1900	109.3	-267.652	-18.6
DMY09D	0.0622	46.7	0.0132	5.2	0.0466	18.5	69.618	4.6
DMY10OND	-0.0187	-21.3	-0.0138	-7.7	-0.0031	-1.8	-261.039	-10.8
Market-making factor								
Quote update	-0.0010	-17.9	-0.0004	-4.3	-0.0005	-4.8	15.132	16.7
dummy for Dec 2009	-0.0043	-46.7	-0.0015	-7.8	-0.0026	-13.8	3.911	3.6
dummy for Oct-Dec 2010	0.0001	2.6	-0.0001	-0.5	0.0001	1.2	9.287	7.0
Trade intensity								
Trade size	0.0019	159.8	0.0011	50.8	0.0007	34.9	7.145	38.4
dummy for Dec 2009	0.0001	8.9	0.0002	8.5	-0.0001	-3.6	2.455	8.2
dummy for Oct-Dec 2010	-0.0004	-21.9	-0.0002	-5.6	-0.0002	-9.1	-0.702	-2.8
Imbalance								
Order imbalance	0.0399	30.9	-0.0064	-2.3	0.0471	16.8	162.098	14.4
dummy for Dec 2009	0.0012	0.6	-0.0022	-0.6	0.0036	0.9	-89.776	-6.8
dummy for Oct-Dec 2010	0.0095	6.5	0.0186	6.0	-0.0093	-3.0	210.452	10.6
Runs Persistency								
Runs Persistency	-0.0014	-7.2	-0.0064	-14.1	0.0050	11.1	-13.094	-5.1
dummy for Dec 2009	-0.0008	-2.7	-0.0024	-4.2	0.0020	3.5	-45.344	-10.6
dummy for Oct-Dec 2010	0.0016	7.6	0.0007	1.4	0.0011	2.3	-14.473	-3.3
TMF1	0.0357	70.1	-0.0138	-14.8	0.0495	52.5	-86.667	-18.9
TMF2	0.0073	17.2	-0.0042	-4.8	0.0110	12.7	-18.699	-4.1
TMF4	0.0017	3.7	0.0073	7.5	-0.0080	-8.3	23.265	4.4
TMF5	-0.0038	-9.2	-0.0085	-10.2	0.0052	6.2	-23.690	-5.2
TMF6	-0.0105	-24.9	0.0080	8.6	-0.0177	-19.2	22.798	4.4
TMF7	-0.0099	-22.2	0.0117	12.6	-0.0212	-23.1	20.620	3.9
TMF8	-0.0096	-23.7	0.0151	18.4	-0.0245	-30.5	14.424	2.8
TMF9	-0.0030	-6.9	0.0245	27.0	-0.0294	-33.3	-8.814	-1.8
TICF	0.0190	61.0	0.0078	14.0	0.0101	17.9	8.030	2.5
LOG of Cap on Nov 2009	-0.0379	-282.4	-0.0043	-16.7	-0.0313	-122.4	-24.325	-9.6
Observation	774,556		728,699		728,699		774,567	
Adjusted R-squared	0.3594		0.0320		0.0740		0.0792	

Note: The t-statistics are White heteroskedasticity-consistent standard errors and covariance.

Figure 1. Effective spreads (daily, tick size).

The daily movement of effective spreads is computed as an average of the effective spreads for stocks belonging to each price range defined in Table 1. Stocks moving in and out of a price range are excluded. The title of the chart indicates the tick size (before and after the introduction of Arrowhead).

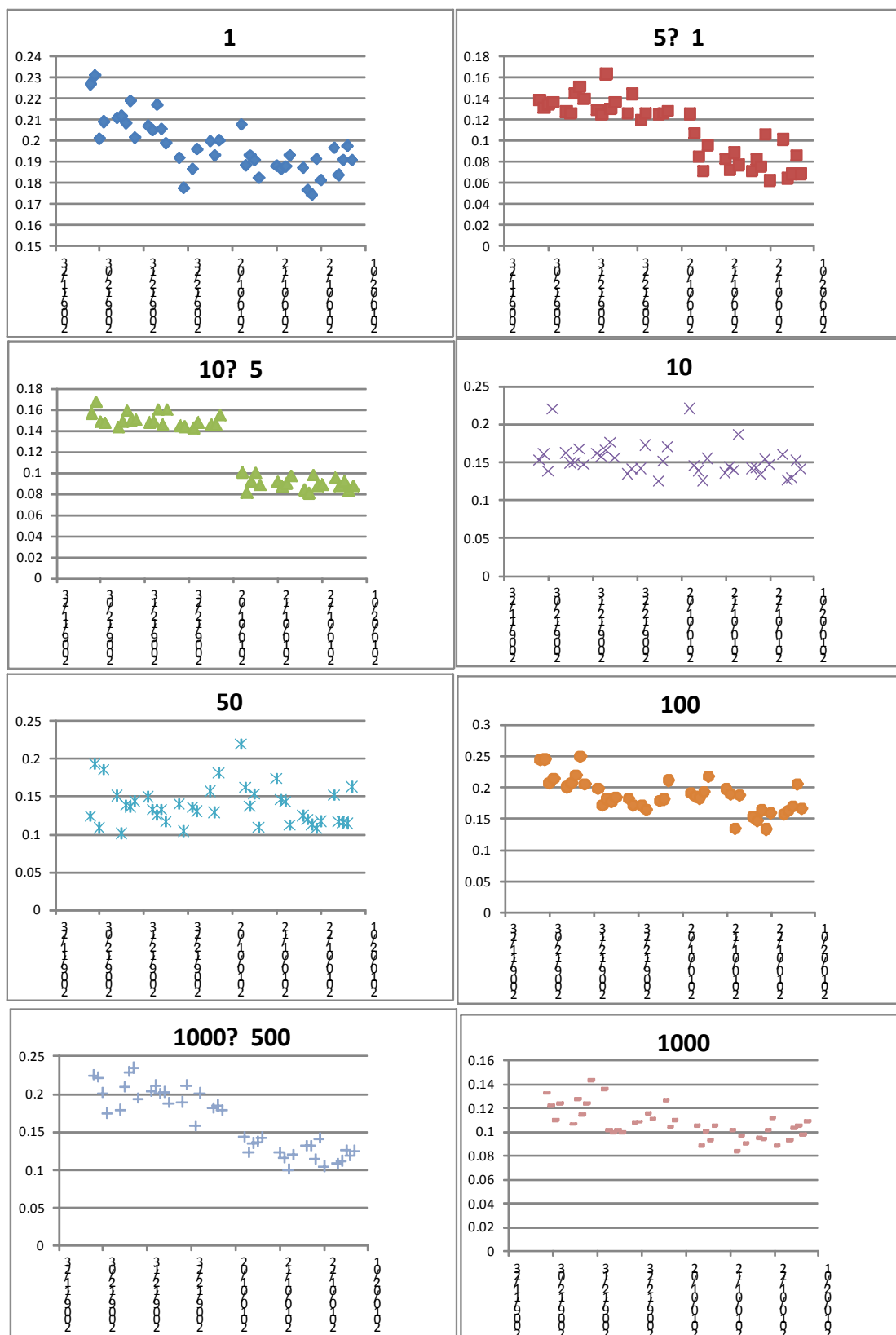


Figure 2. Changes in depth (percentage).

Depth is computed as an average of the best asks and bids for individual stocks. We then compare the depths for December 2009 and January 2010.

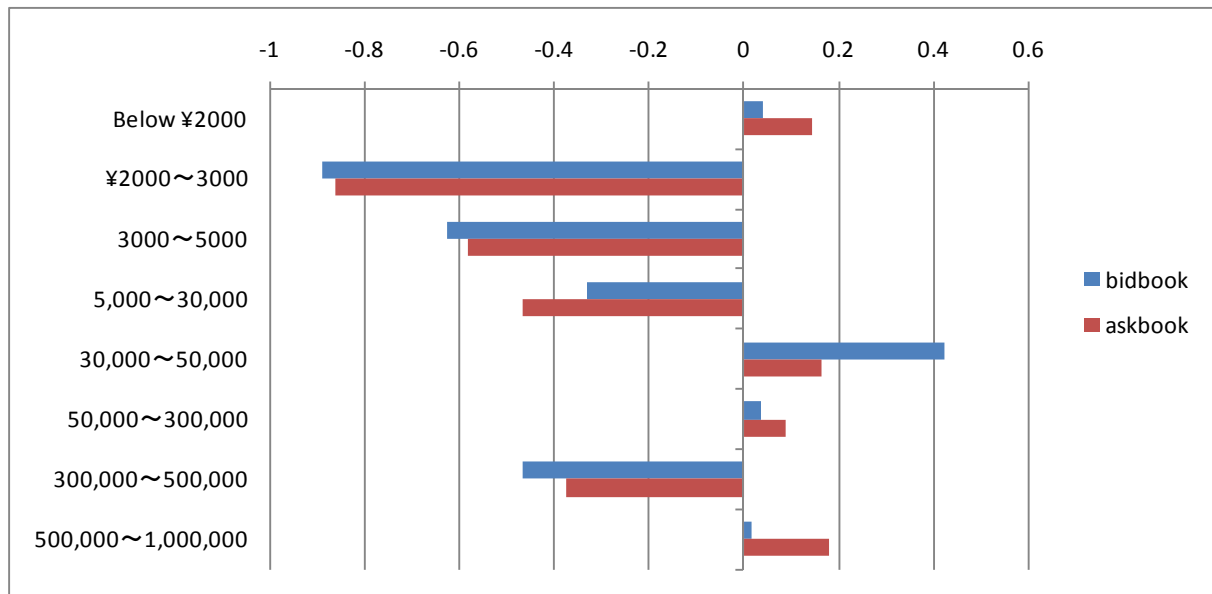
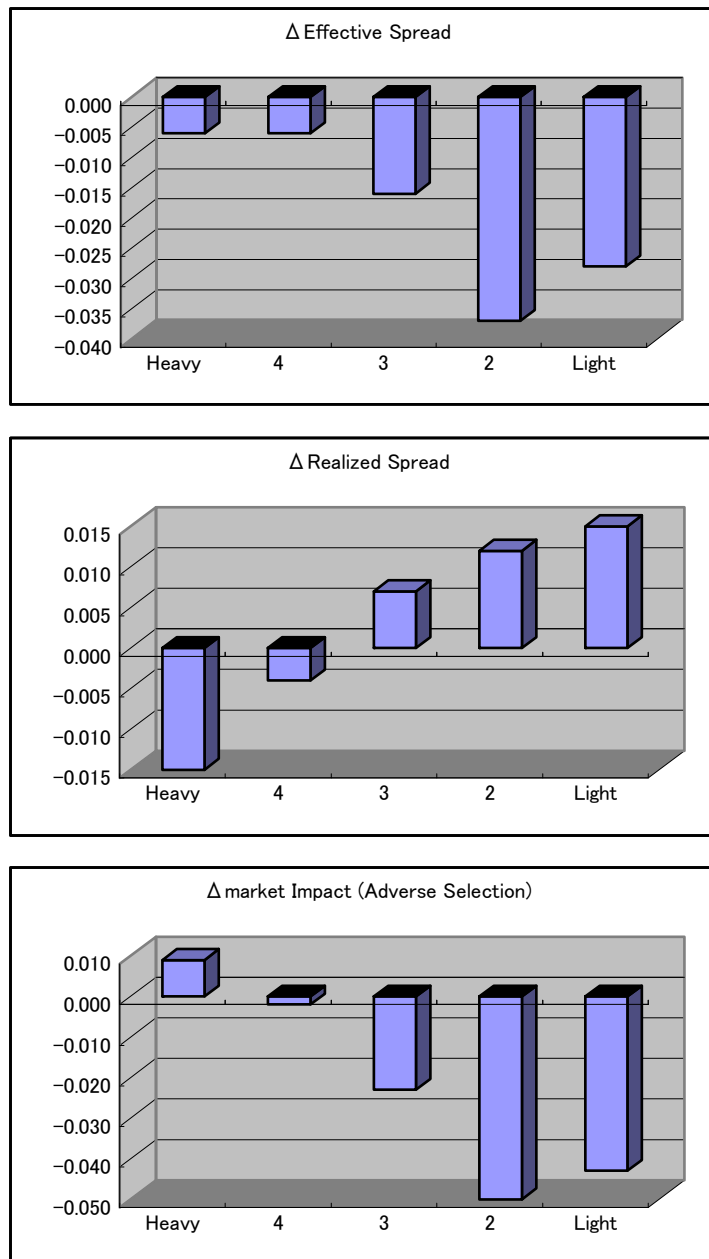


Figure 3. Spread changes by HFT effect quintiles.

Panel A: Stocks unaffected by changes in tick size. The bar charts indicate differences in effective spreads, realized spreads, and market impact in Panel A of Table 5.



Panel B: Stocks affected by changes in tick size. The bar charts indicate differences in effective spreads, realized spreads, and market impact in Panel B of Table 5.

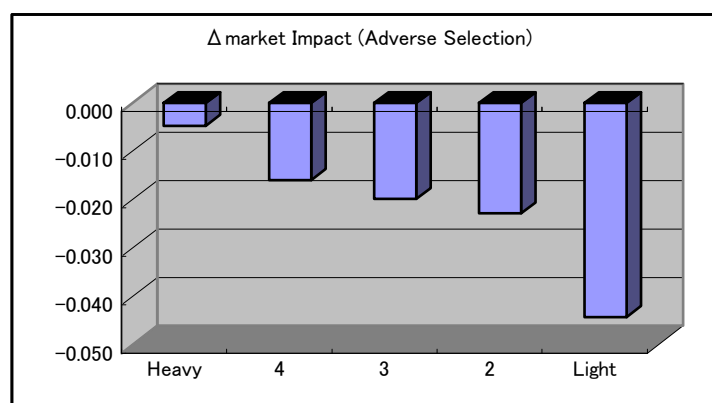
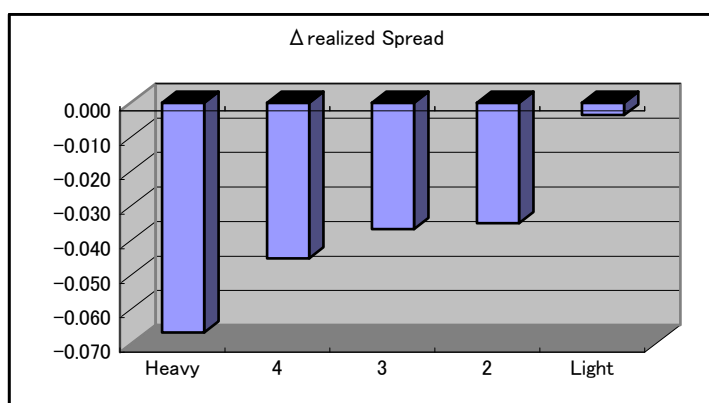
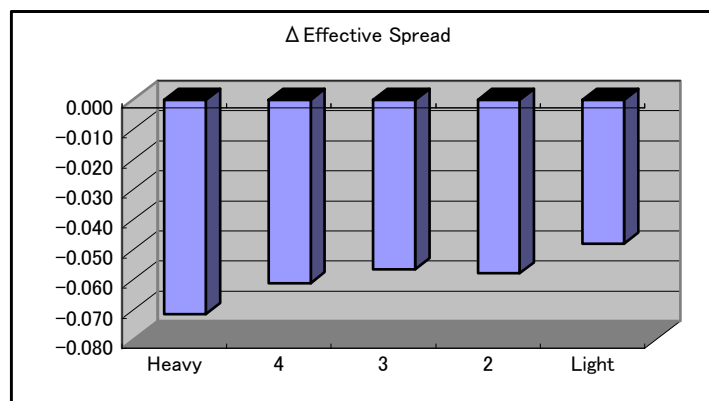


Figure 4. Decomposed effects of market making proxy versus three order flow factors. The changes brought about by the new trading system are decomposed into four components: market making (quote update) and order flow intensity, that is, trade size, order imbalance, and the persistence of runs. Each amount is computed by multiplying the estimated coefficients of the variables and the mean values of the explanatory variables for each period.

